

App Organization

Section Review

Python Packages

- Create a **`__init__.py`** file inside a directory so that Python can identify it as a package.
- We can references packages in the import code.
 - **`import package.my_file`**
- We created packages for our **app**, **dependencies**, **routers**, and **models** folders.

The APIRouter Class

- The **APIRouter** class is a container for related routes.
- The router supports the same methods (**get/post**, etc.) as the **FastAPI** object.
- Routers help split up routes across multiple files.

Adding the Routers

- Import the **router** objects from their respective files/modules.
- Invoke the **include_router** method on the main FastAPI application object and pass in the router.
- The order of the **include_router** calls determines the order that routes are added.

Configuring the Routers

- The **APIRouter** constructor accepts a **prefix** parameter to set a fixed path prefix before each route.
- The **tags** parameter adds a tag to all routes within the router. Swagger uses tags to group routes.
- A router can configure dependencies. The router can invoke a function or instantiate a class before any route handler executes.

Global Dependencies

- The **FastAPI** constructor accepts its own **dependencies** list.
- Pass in the **Depends** function with the dependency function.
- FastAPI will run the dependency once before every request.

Dependencies in Route Decorators

- The route methods (**get/post** etc) accept a **dependencies** list as well.
- The advantage is that the we do not need to declare the dependency as an unused function parameter.
- An alternate option is to prefix the parameter with **_** to indicate it is not used by the function body.

Patterns of Code Organization

- A **service** is a class/function that handles the execution logic for an operation. It pulls business logic out of route handlers.
- The BetaAcid codebase uses a pattern of routers -> service classes -> database client classes.
- The Netflix Dispatch codebase uses a pattern of routers -> service functions.